AMRITA SCHOOL OF ENGINEERING, BENGALURU

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SYSTEM DESIGN FOR 23ECE383 VLSI DESIGN LABORATORY

REPORT

JULY TO NOVEMBER 2026

Implementation of a Compact Multiphase Divide-by-Two CMOS Frequency Divider in 45 nm CMOS (Cadence GPDK045)

Submitted by

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TITLE OF THE SYSTEM DESIGN PROJECT:

Implementation of a Compact Multiphase Divide-by-Two CMOS Frequency Divider in 45 nm CMOS (Cadence GPDK045)

TITLE OF THE PAPER:

A Compact Multiphase Divide-by-Two CMOS Frequency Divider

DETAILS OF THE PAPER (SOURCE OF THE TITLE):

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CIRCUIT DIAGRAM:

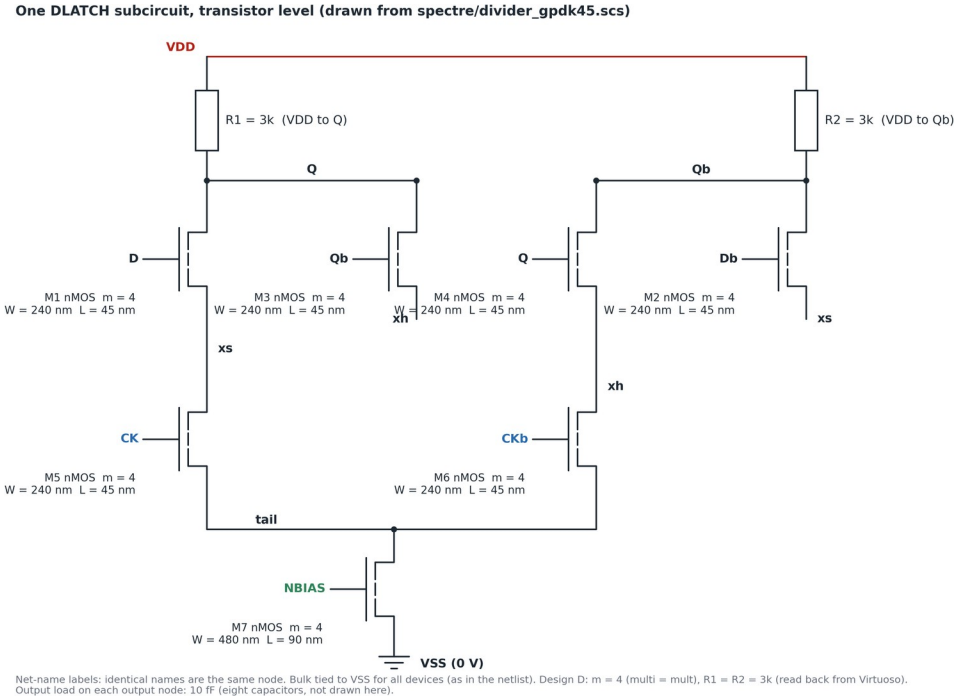


Figure 1: Schematic of the differential DLATCH stage (M1-M7, R1, R2). Sizing: M1-M6 W = 240 nm, L = 45 nm; M7 W = 480 nm, L = 90 nm. Values shown for design D (m = 4, R1 = R2 = 3 kOhm; baseline uses m = 1, R = 8 kOhm).

THEORY AND WORKING PRINCIPLE:

Topology. The divider consists of four differential D-latches in a ring with alternating cross-connections (Q1b-D2, Q2-D3, Q3b-D4, Q4-D1b) and inverted clocks on latches 2 and 3 (Kuo et al.). Each latch contains sampling pair M1-M2, clock switches M5-M6, regenerative hold pair M3-M4, tail transistor M7, and loads R1-R2. Sampling occurs when CK is high, and regeneration occurs when CK is low, producing eight phases at half the input frequency.

Elmore delay model. Latch propagation delay is modeled as t_pd = (8C)(R/2) = 4RC, where each output carries effective capacitance 8C and load R/2 = R1. Division requires T_CK/2 > 2 Td. For load capacitance C_L = 10 fF, the parasitic-free lower bound is Td >= R1 C_L, yielding 30 ps for design D (R1 = 3 kOhm) and 20 ps for design B (R1 = 2 kOhm). At 1.4 GHz (design D), 2 Td >= 60 ps satisfies T_CK/2 = 357 ps. At 6 GHz (design B), 2 Td >= 40 ps satisfies T_CK/2 = 83 ps. Design B divides up to 6 GHz and fails at 6.2 GHz (T_CK/2 = 80.6 ps). Its measured CK-to-Q1 delay of 66.10 ps is 3.3x the intrinsic lower bound (79% of T_CK/2), showing that parasitic capacitances dominate near the frequency limit.

Stack resistance and regeneration. The pull-down network forms a three-transistor stack (M1, M5, M7) with effective resistance 3 Ron/m. Latch regeneration requires gm R1 > 1, yielding R1 >= 1.1 x (3 Ron/m). With Ron = 2.3 kOhm, the unit stack is 6.90 kOhm, requiring R1 >= 7.59 kOhm. The paper load (1.15 kOhm) falls far below this threshold, preventing division. Measured thresholds for m = 1 span 6 to 7 kOhm at vb = 1 V, and 7 to 8 kOhm at vb = 0.8 V. For m = 4, the required threshold is 1.90 kOhm (band 1.88 to 2.25 kOhm); design D (R1 = 3 kOhm) provides margin, while design B (R1 = 2 kOhm) operates within the marginal band. Measured division agrees with the rule for 5 of 5 tested parameter pairs. The unit device aspect ratio W/L = 5.33 (240 nm / 45 nm) matches the reference design ratio of 5.00 (0.9 um / 180 nm).

Phase-step relationship. Outputs toggle in pairs separated by t_pair (measured between rising edges of Q2 and Q3b), yielding phase step a = 180 t_pair f_in degrees, alternating with 90 - a degrees. For design D at 1.4 GHz, t_pair = 25.95 ps gives steps of 83.46 and 6.54 deg (a = 180 x 25.95 ps x 1.4 GHz = 6.54 deg). For design B at 6 GHz, t_pair = 38.16 ps gives 48.78 and 41.22 deg (a = 41.22 deg). Baseline U8k at 1.4 GHz gives 71.42 and 18.59 deg (t_pair = 73.75 ps, a = 18.59 deg). Nominal 45 deg spacing requires T_CK = 4 t_pair: 103.8 ps (9.63 GHz) for D and 152.7 ps (6.55 GHz) for B. Phase error decreases as frequency increases toward the divider limit, matching the reference paper value (3.75 deg) only in design B at 6 GHz.

Definitions of performance metrics. Delay is the average time from each CK rising edge to the next transition on any output (25.99 ps for design D; single-edge CK-to-Q1 delay is 51.95 ps). Power is VDD times the average supply current over full cycles. PDP is power times delay (fJ). Division is valid when all eight outputs toggle at f_in/2 within 0.1 percent with peak-to-peak swing >= 0.1 V.

TOOL & TECHNOLOGY USED:

Tools and environment: Cadence Virtuoso IC615 (6.1.5), Spectre circuit simulator (MMSIM 12.1), and Assura physical verification (4.1). Process technology is Cadence GPDK045 v6.0 (45 nm generic CMOS), using standard-threshold NMOS devices (g45n1svt, 1.0 V nominal) evaluated across tt, ff, ss, fs, and sf corners.

Device sizing: unit transistors M1-M6 have W = 240 nm, L = 45 nm, and tail transistor M7 has W = 480 nm, L = 90 nm. Device multiplier m scales channel widths in parallel.

Operating conditions: nominal supply VDD = 1 V (PVT range 0.9 V to 1.1 V), nominal temperature 27 C, clock transition times 20 ps, tail bias vb = 1 V, and load capacitance 10 fF per output. Simulation dataset: "Transient Analysis `tran': time = (0 s -> 40 ns)" (12:25:58 PM, Sat Oct 31, 2026).

Virtuoso schematic parameter extraction for the design D DLATCH cell: R1 and R2 = 3 kOhm; M1-M6 w = 240n, l = 45n, m = 4; M7 w = 480n, l = 90n, m = 4. These match Table 2 and the nominal transistor dimensions.

Circuit schematics for the DLATCH and ring architecture are illustrated in Figure 1 and Figure 8.

SCREENSHOT OF THE SIMULATOR BASED CIRCUIT IMPLEMENTATION:

Baseline circuit without sizing (U8k): unit devices (m = 1), load R1 = R2 = 8 kOhm, tail bias vb = 1 V, and input 1.4 GHz (tt, 27 C, VDD = 1 V, g_1.4G_8k_1.0_10f_m1). All eight outputs divide to 700.0 MHz, with Q1 swinging from 0.749 V to 0.978 V.

Selection of R: load resistance was swept from 5 to 12 kOhm (12 runs) at vb = 0.8 V and 1 V. Division fails up to 6 kOhm and succeeds from 7 kOhm at vb = 1 V; at vb = 0.8 V it fails up to 7 kOhm and divides from 8 kOhm. The value R = 8 kOhm at vb = 1 V was selected as the minimum round value dividing at both biases.

Failure of paper sizing: direct implementation of the paper rule (m = 1, R1 = R2 = Ron/2 = 1.15 kOhm, Ron = 2.3 kOhm) fails to divide at 1.4 GHz. Output Q1 oscillates at 2799.8 MHz (2.0 x f_in) with only 15.3 mV swing (Figure 9). Supply power is 0.165 mW; delay, PDP, and phase error are undefined.

Mechanism: the pull-down path comprises three stacked devices (M1, M5, M7) with effective resistance 3 Ron = 6.90 kOhm for m = 1. Regeneration requires R1 >= 1.1 x 3 Ron = 7.59 kOhm, consistent with simulated bounds of 6-7 kOhm at vb = 1 V and 7-8 kOhm at vb = 0.8 V.

Table 1: Device list, circuit without the sizing concept

Device	W	L	Multiplier	Role
M1, M2	240 nm	45 nm	1	sampling pair (differential inputs)
M3, M4	240 nm	45 nm	1	hold pair (cross-coupled regenerative pair)
M5	240 nm	45 nm	1	switch controlled by CK
M6	240 nm	45 nm	1	switch controlled by CKb
M7	480 nm	90 nm	1	tail current source, gate at vb = 1 V
R1, R2	n/a	n/a	n/a	load resistors, 8 kOhm each

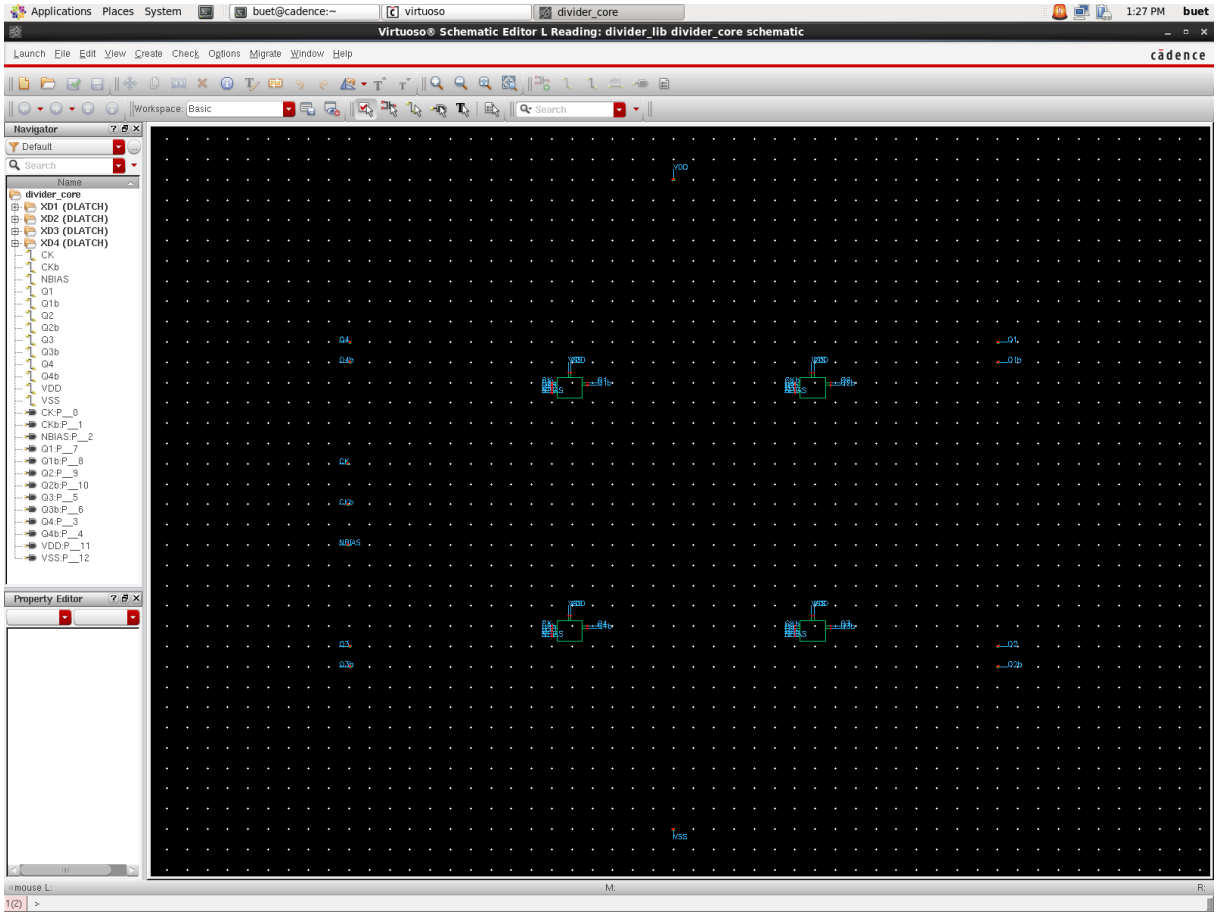


Figure 2: Virtuoso schematic editor window showing the divider core with four DLATCH instances (XD1 to XD4) connected in a ring. Baseline and design D share this topology, differing only in m and R (Tables 1 and 2). Schematic check: DLATCH 0/0, divider_core 0/0, tb_divider 0/1 errors/warnings.

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR (DC AND TRANSIENT):

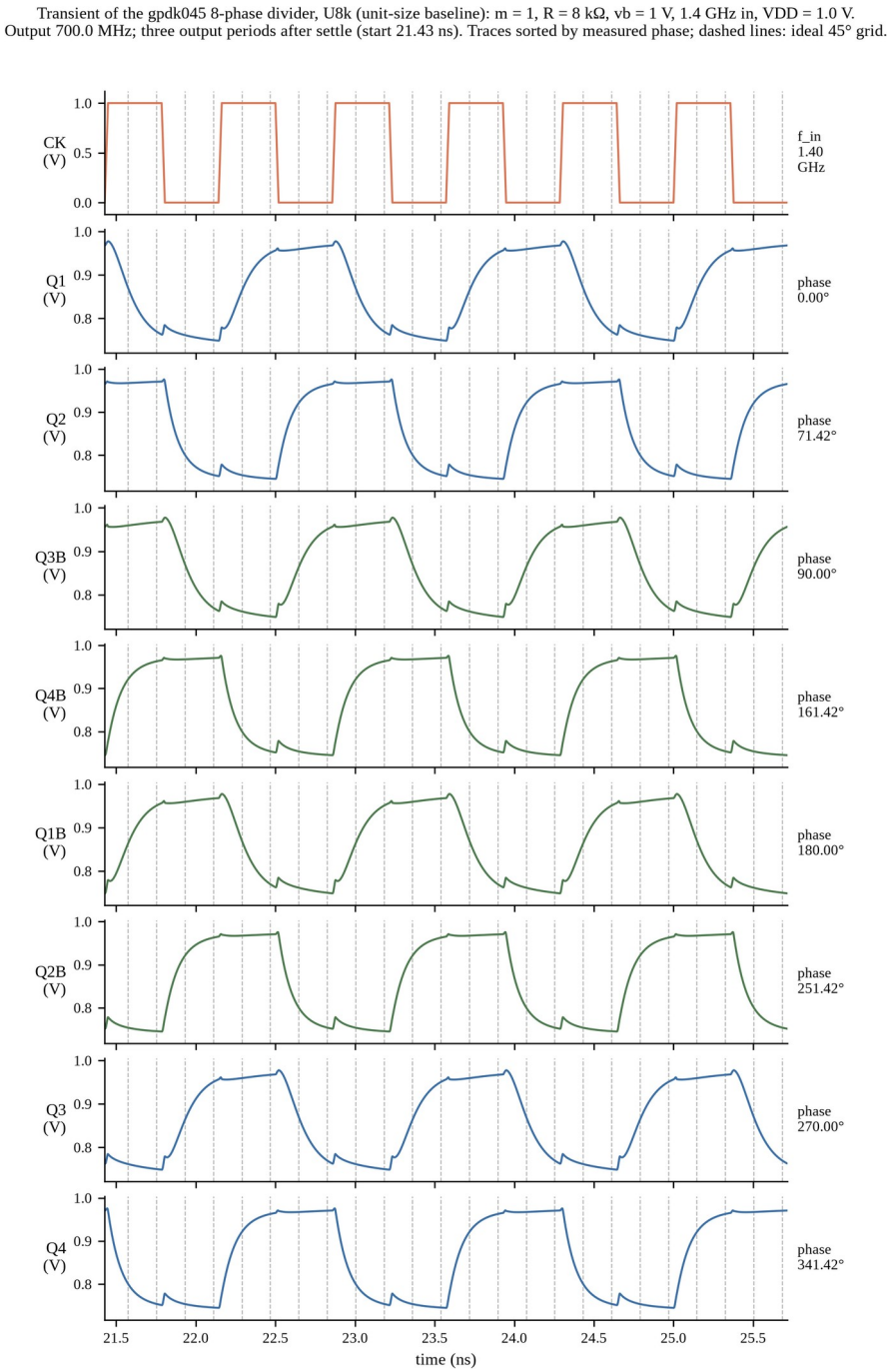


Figure 3: Transient waveforms for the baseline divider without sizing (m = 1, R1 = R2 = 8 kOhm, vb = 1 V) at 1.4 GHz. All eight outputs divide to 700.0 MHz, with Q1 swinging from 0.749 V to 0.978 V. Waveforms for paper sizing appear in Figure 9.

Simulation driven directly from the Virtuoso schematic (OCEAN)

To evaluate the sized divider circuit (design D) directly from the Cadence schematic environment, the testbench schematic was simulated with Spectre at the nominal typical corner (tt, 27 C, 1.0 V supply, 1.0 V bias) with a 1.4 GHz input clock. The circuit uses load resistors R1 = R2 = 3 kOhm and transistor multiplicity m = 4. All 8 outputs divide to 700 MHz with a frequency division ratio of 2.0.

Propagation delay from the clock to the divider outputs is 26.66 ps, compared with 25.99 ps from the standalone netlist. Average supply power is 0.455 mW, compared with 0.541 mW from the standalone netlist run. The power differs by about 16 percent between the two flows. The cause is under investigation; the averaging window and the device parameters added by the Virtuoso netlister differ; the standalone value is used for the power and PDP tables.

The testbench schematic is shown in Figure 20.

SCREENSHOT OF THE CIRCUIT BASED ON SIZING CONCEPT:

Sized divider (design D): device multiplier m = 4, load R1 = R2 = 3 kOhm, and input 1.4 GHz. The pull-down stack resistance is 3 Ron/m = 1.72 kOhm, requiring load resistance of 1.90 kOhm (tolerance band 1.88 to 2.25 kOhm). Load R = 3 kOhm provides adequate margin. The same sizing rule applies to design B at 6 GHz (m = 4, R1 = R2 = 2 kOhm). Transistor M7 operates in triode, with transient drain voltage remaining below Vdsat.

Comparison with baseline (Table 3): sizing cuts delay from 62.21 ps (U8k) to 25.99 ps (D) (2.39x) and increases swing from 0.2289 V to 0.4344 V. Power rises from 0.1375 mW to 0.5413 mW (3.94x), PDP from 8.557 fJ to 14.067 fJ (1.64x), and phase-step error from 26.42 deg to 38.46 deg at 1.4 GHz.

Table 2: Device list, circuit based on the sizing concept

Device	W	L	Multiplier	Role
M1, M2	240 nm	45 nm	4	sampling pair (differential inputs)
M3, M4	240 nm	45 nm	4	hold pair (cross-coupled regenerative pair)
M5	240 nm	45 nm	4	switch controlled by CK
M6	240 nm	45 nm	4	switch controlled by CKb
M7	480 nm	90 nm	4	tail current source, gate at vb = 1 V
R1, R2	n/a	n/a	n/a	load resistors, 3 kOhm each

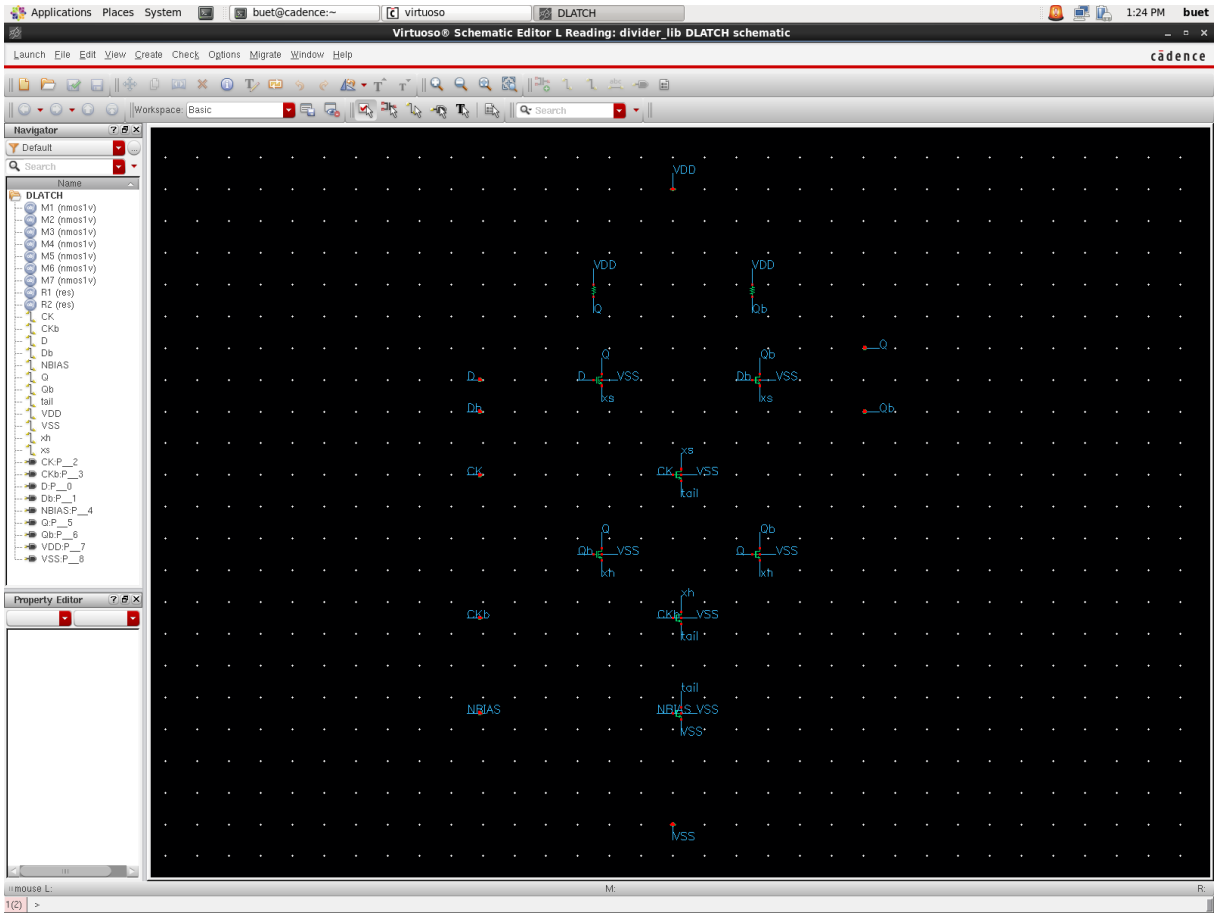


Figure 4: Virtuoso schematic editor window showing the DLATCH cell with sizing applied (design D: m = 4, R1 = R2 = 3 kOhm), consistent with Table 2. Schematic check: DLATCH 0/0, divider_core 0/0, tb_divider 0/1 errors/warnings.

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR FOR THE CIRCUIT WITH SIZING CONCEPT (DC AND TRANSIENT):

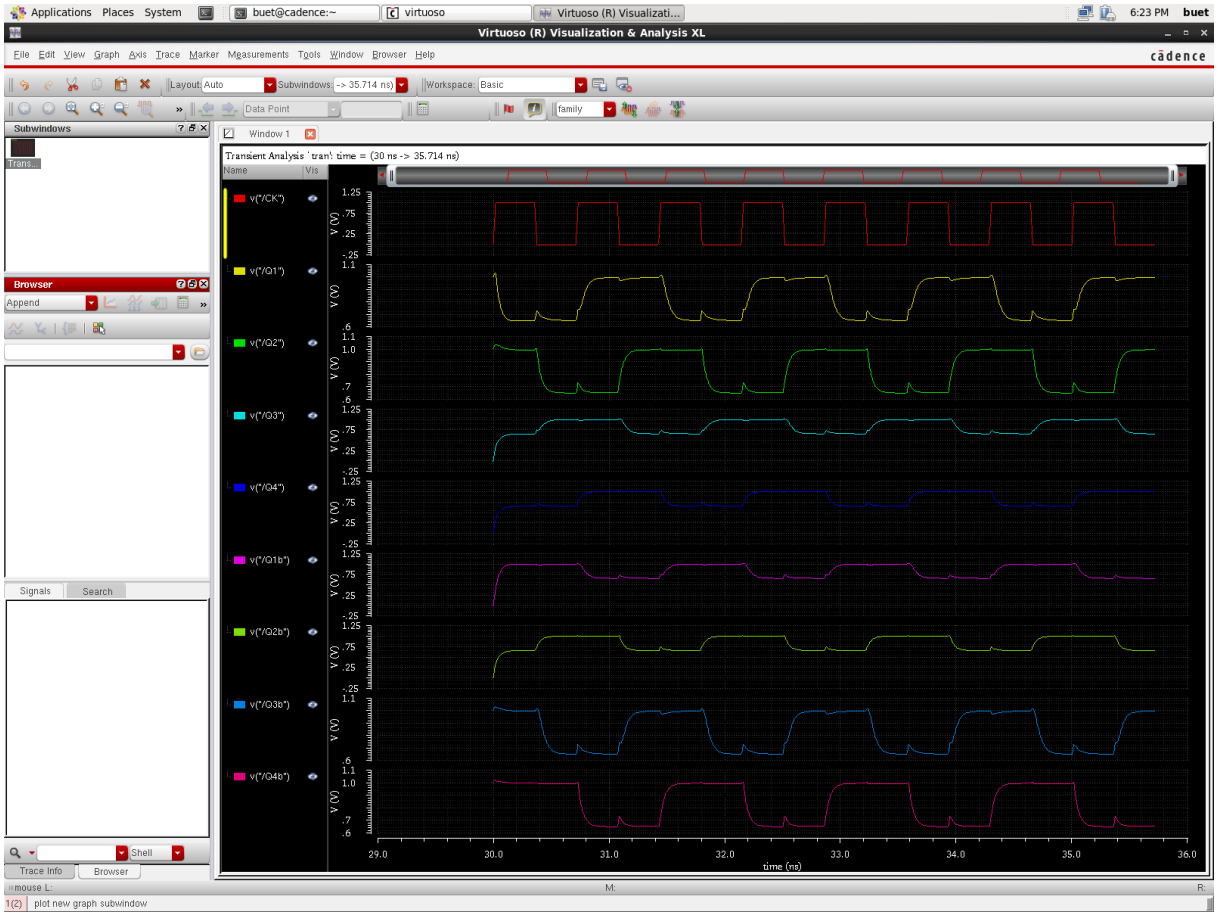


Figure 5: ViVA transient waveforms of design D from the Virtuoso testbench via Spectre.

PROVIDE THE VALUES FOR THE FOLLOWING ALONG WITH THE SCREENSHOTS AS PROOF:

Table 3: Performance values with and without the sizing concept, with the proof figures for each row

Parameter	Value without Sizing	Value with Sizing	Proof (figure)
Power calculation (P)	0.1375 mW	0.5413 mW	Fig. 15 (supply current, both columns)
Delay calculation (t_pd)	62.21 ps	25.99 ps	Calculated mean delay (see note)
Power Delay Product (PDP)	8.557 fJ	14.067 fJ	Calculated: power x delay
Input frequency (f_in)	1.4 GHz	1.4 GHz	Fig. 3 (without), Fig. 5 (with)
Output frequency (f_out)	700.0 MHz	700.0 MHz	Fig. 3 (without), Fig. 5 (with)
Max phase-step error	26.42 deg	38.46 deg	Fig. 11
Output swing	0.2289 V	0.4344 V	Fig. 3 (without), Fig. 5 (with)
Tail-device saturation check (M7)	**Vds = 0.0671 V, Vdsat = 0.3578 V: NOT saturated (triode)**	**Vds = 0.0669 V, Vdsat = 0.3578 V: NOT saturated (triode)**	Fig. 12 (without), Fig. 13 (with)

Notes: Baseline without sizing corresponds to unit devices U8k (m = 1, R = 8 kOhm, vb = 1 V). The paper sizing circuit (m = 1, R = 1.15 kOhm) fails to divide (supply power 0.165 mW; delay, PDP, and phase error undefined; Figure 9). Delay is the average time from CK rising edges to the next output transition (first-edge CK-to-Q1 delay: 135.96 ps for U8k, 51.95 ps for D). Vds indicates peak transient tail voltage V(XD1.tail); Vdsat is the extracted saturation voltage for M7.

SCREENSHOT OF THE PVT ANALYSIS:

Table 4: PVT summary for design D at 1.4 GHz (41 of 45 pass; FAIL at 125 °C, 0.9 V for tt, ss, fs, sf) (41 of 45 corners pass)

Corner	Temp (°C)	0.9 V	1.0 V	1.1 V
tt	-40	PASS	PASS	PASS
tt	27	PASS	PASS	PASS
tt	125	**FAIL**	PASS	PASS
ff	-40	PASS	PASS	PASS
ff	27	PASS	PASS	PASS
ff	125	PASS	PASS	PASS
ss	-40	PASS	PASS	PASS
ss	27	PASS	PASS	PASS
ss	125	**FAIL**	PASS	PASS
fs	-40	PASS	PASS	PASS
fs	27	PASS	PASS	PASS
fs	125	**FAIL**	PASS	PASS
sf	-40	PASS	PASS	PASS
sf	27	PASS	PASS	PASS
sf	125	**FAIL**	PASS	PASS

Table 5: PVT summary for design B at 6.0 GHz (21 of 45 pass) (21 of 45 corners pass)

Corner	Temp (°C)	0.9 V	1.0 V	1.1 V
tt	-40	PASS	PASS	PASS
tt	27	**FAIL**	PASS	PASS
tt	125	**FAIL**	**FAIL**	**FAIL**
ff	-40	PASS	PASS	PASS
ff	27	**FAIL**	PASS	PASS
ff	125	**FAIL**	**FAIL**	**FAIL**
ss	-40	**FAIL**	PASS	PASS
ss	27	**FAIL**	**FAIL**	PASS
ss	125	**FAIL**	**FAIL**	**FAIL**
fs	-40	PASS	PASS	PASS
fs	27	**FAIL**	PASS	PASS
fs	125	**FAIL**	**FAIL**	**FAIL**
sf	-40	**FAIL**	PASS	PASS
sf	27	**FAIL**	**FAIL**	PASS
sf	125	**FAIL**	**FAIL**	**FAIL**

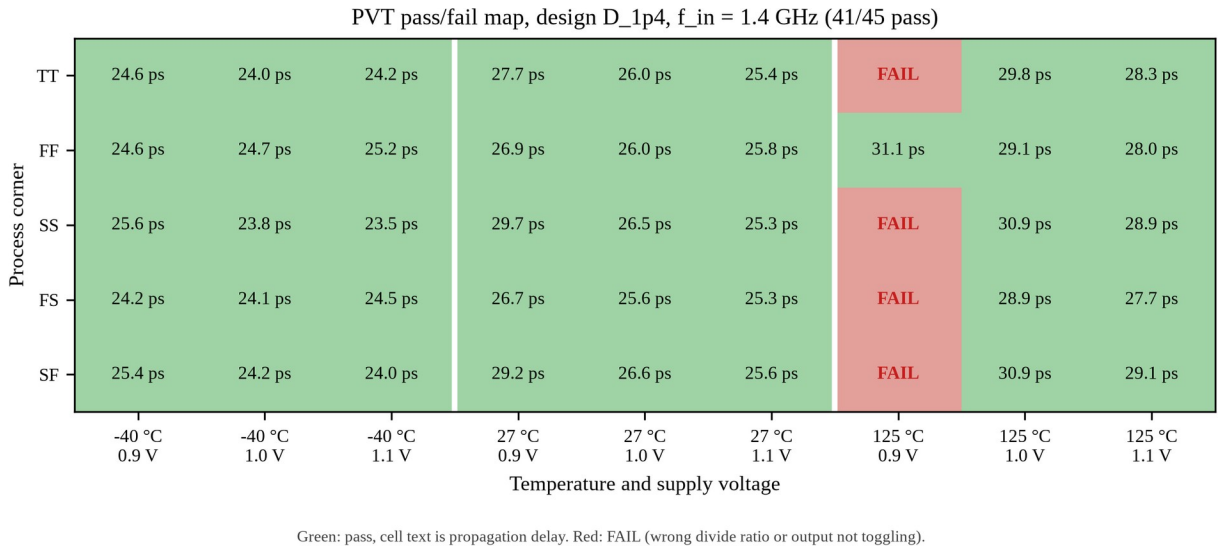


Figure 6: PVT pass/fail map for design D at 1.4 GHz across 5 corners, temperatures -40/27/125 C, and supplies 0.9/1/1.1 V (41 of 45 points pass).

SCREENSHOT OF THE LAYOUT:

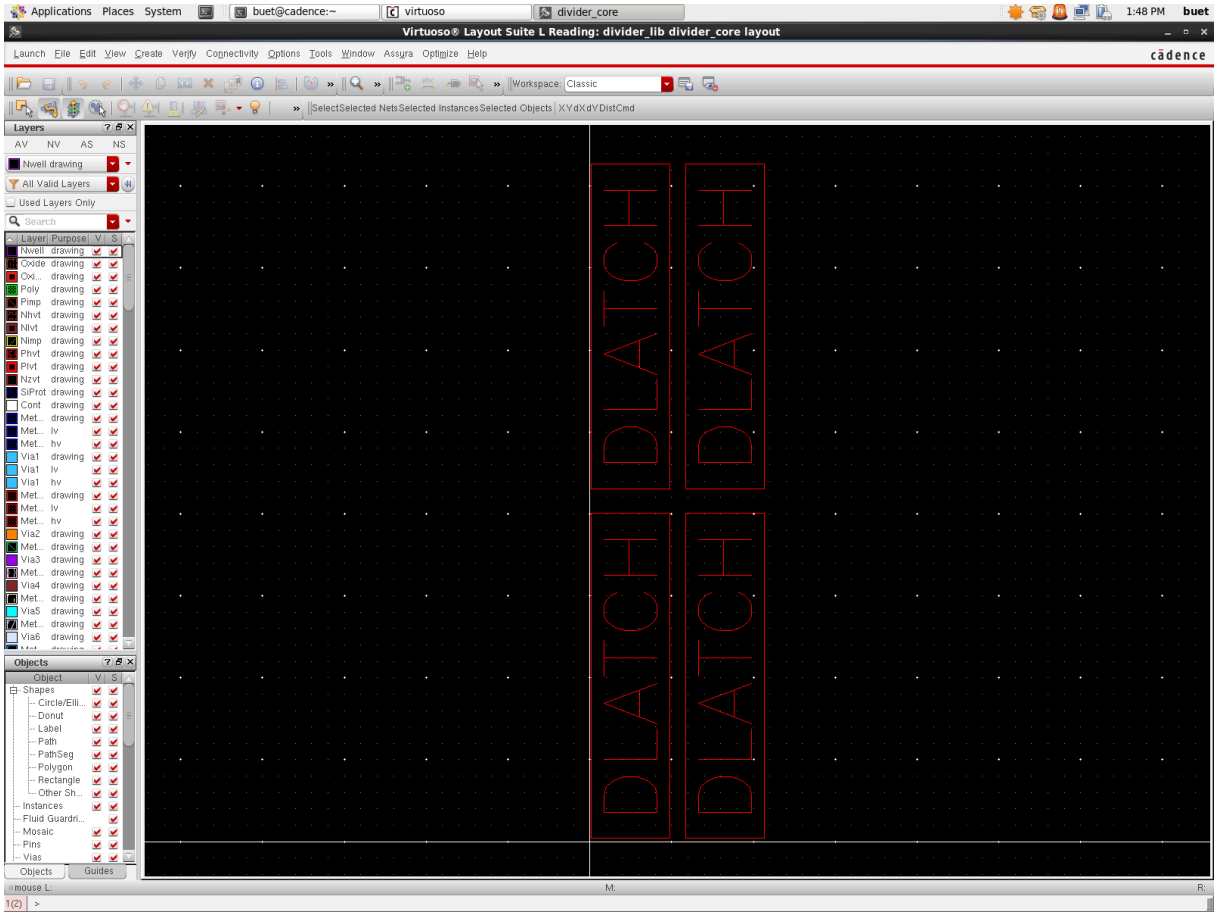


Figure 7: Virtuoso layout editor showing divider_core with four DLATCH instances (XD1 to XD4) placed in a 2x2 array (placement only, unrouted).

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR FOR THE LAYOUT:

The layout is placement only: it is not routed, and LVS and post-layout simulation were not run.

ANY OTHER RELEVANT INFORMATION:

Layout. The DLATCH cell contains 7 unit nMOS pcells (0.595 um wide) and 2 poly resistors (segW 0.8 um, segL 7.855 um, 200 ohm/sq, target 2000 ohm). Transistors M1-M6 use 2 fingers of 120 nm (240 nm, L = 45 nm); M7 uses 4 fingers of 120 nm (480 nm, L = 90 nm) for m = 1. Design D requires 4x width (m = 4) and 3 kOhm load (baseline uses 8 kOhm). DLATCH boundary is 4.875 x 20.28 um. Cell divider_core places XD1 R0, XD2 MY, XD3 MX, XD4 R180 with a 1 um gap (clock-star length 98.62 um each). The layout is placement only: it is not routed, and LVS and post-layout simulation were not run. An automatic routing run with Cadence's Chip Assembly Router completed 136 of 137 connections but produced 134 DRC

errors and mismatches in LVS, so the placement layout is the one reported. Cell layouts are shown in Figure 7 (divider_core) and Figure 22 (DLATCH).

DRC verification. Assura DRC on DLATCH reports 21 errors across 3 rules (Figure 23). Rules 220 (1) and 221 (6) give 7 substrate-stamp errors from unrouted taps at device pitch 0.755 um. Rule 595 (14) gives 14 Metal1 minimum-area errors (threshold 0.02 um^2); 4 markers measure 0.06 x 0.12 um (0.0072 um^2) across x = 0.96-1.155 um, y = 1-2.02 um. Contact pins (0.2 x 0.15 um = 0.03 um^2) meet the minimum area. The unrouted layout is not DRC clean.

Schematic connectivity. Pin-by-pin comparison of the 4 DLATCH instances (XD1-XD4) against the netlist confirmed full consistency with 0 mismatches. Schematic check warnings: DLATCH 0/0, divider_core 0/0, and tb_divider 0/1.

Design B operating point. At 6 GHz (m = 4, R1 = R2 = 2 kOhm), the divider achieves 3000.0 MHz output, 0.4929 mW power, 27.94 ps delay, 13.768 fJ PDP, and 3.78 deg maximum phase error (48.78 and 41.22 deg steps). Output Q1 swings 0.135 V (0.818 V to 0.953 V), above the 0.1 V limit. Peak tail voltage reaches 66.2 mV, below Vdsat = 357.8 mV (triode, Figure 14). PVT results appear in Table 4.

DC operating points. At t = 0 (clocks low, switches off), tail current is negligible (Vgs = 1 V). For paper sizing, M7 is in triode (Vds = 56.3 nV, Ids = 35.6 pA). For design D, M7 is in triode (Vds = 54.4 nV, Ids = 137.4 pA). For design B (m = 4, vb = 1 V), M7 is in triode (Vds = 54.4 nV, Ids = 137.4 pA).

Tail transistor operating region. With Vgs = 1 V and Vth = 527.5 mV, extracted Vdsat is 357.8 mV (at t = 0, gm = 200.9 pS, gds = 2.531 mS, Vds = 54 nV). In transient operation, V(XD1.tail) spans 8.1 to 66.9 mV for design D (XD2 reaches 68.2 mV) and 8.0 to 67.1 mV for baseline U8k. Because Vds remains below Vdsat, M7 operates in triode rather than saturation (Figures 12 and 13).

Frequency sweeps. Design B (m = 4, R = 2 kOhm) was swept from 1.4 to 6.2 GHz: it divides at 1.4, 2, 3, 4, 5, 5.5, 5.8, 6 GHz and fails at 6.2 GHz. Design D (m = 4, R = 3 kOhm) was swept from 3 to 8 GHz: it divides at 3, 4, 5, 5.5 GHz and fails at 6, 6.5, 7, 8 GHz (the 1.4 GHz operating point also divides). Design E (m = 6, R = 2 kOhm) was swept from 3 to 8 GHz: it divides at 3, 4, 5, 5.5, 6, 6.5, 7 GHz and fails at 8 GHz (1.4 GHz divides at tt, 27 C, 1.0 V). Maximum phase-step error at the highest dividing frequency is 3.78 deg for B, 13.82 deg for D, and 15.24 deg for E, compared with 38.68, 30.89, and 34.32 deg at the lower frequency bounds. In all cases, phase error improves as frequency increases (Figure 16).

Parameter exploration (16 points): m = 4, R = 2.2 kOhm divides up to 6.25 GHz (5.57 deg phase-step error there); m = 4, R = 2.5 kOhm divides up to 6 GHz (11.23 deg phase-step error there) and fails at 6.25 GHz; m = 5, R = 2.2 kOhm divides up to 6.25 GHz (16.56 deg phase-step error there); m = 5, R = 2.5 kOhm divides up to 6.25 GHz (14.36 deg phase-step error there). These nominal simulations characterize the upper operating boundary near the frequency limit of design B.

PVT analysis. Across PVT corners, design D at 1.4 GHz achieved: 41 of 45 pass; 4 fail (125 C: 4): tt_125_0.9, ss_125_0.9, fs_125_0.9, sf_125_0.9. At 1.4 GHz, design B yielded: 27 of 45 pass; 18 fail (27 C: 3, 125 C: 15): tt_27_0.9, tt_125_0.9, tt_125_1.0, tt_125_1.1, ff_125_0.9, ff_125_1.0, ff_125_1.1, ss_27_0.9, ss_125_0.9, ss_125_1.0, ss_125_1.1, fs_125_0.9, fs_125_1.0, fs_125_1.1, sf_27_0.9, sf_125_0.9, sf_125_1.0, sf_125_1.1, and design E (m = 6, R = 2 kOhm) yielded: 41 of 45 pass; 4 fail (125 C: 4): tt_125_0.9, ss_125_0.9, fs_125_0.9, sf_125_0.9. At 6 GHz, design B achieved: 21 of 45 pass; 24 fail (-40 C: 2, 27 C: 7, 125 C: 15): tt_27_0.9, tt_125_0.9, tt_125_1.0, tt_125_1.1, ff_27_0.9, ff_125_0.9, ff_125_1.0, ff_125_1.1, ss_-40_0.9, ss_27_0.9, ss_27_1.0, ss_125_0.9, ss_125_1.0, ss_125_1.1, fs_27_0.9, fs_125_0.9, fs_125_1.0, fs_125_1.1, sf_-40_0.9, sf_27_0.9, sf_27_1.0, sf_125_0.9, sf_125_1.0, sf_125_1.1. Passing corners of design D span 0.246-1.078 mW and 23.54-31.05 ps. In Table 4, ratio is f_in/f_out (ratio 2 denotes correct division; 0.5 denotes outputs at 2 f_in).

Transient delay and edge times. First-edge CK-to-Q1 delays are 135.96 ps (U8k), 51.95 ps (design D), and 66.10 ps (design B). Multi-phase average delay for design D is 25.99 ps. Output Q1 rise and fall times are 134.6 ps and 102.6 ps.

Supply current. Design D average supply current is 0.54144 mA (0.541337 mA integrated from transient, matching within 0.018%). Peak current is 0.560 mA (Figure 15).

Comparison with reference literature. Reported values from Kuo et al. (TSMC 180 nm, 1.8 V, 1.4 GHz) are 2.34 mW, phase error within 3.75 deg, and core area 91.7 x 52.9 um^2. In 45 nm, design D at 1.4 GHz consumes 0.5413 mW with 38.46 deg phase error; design B reaches 3.78 deg at 6 GHz.

Figure index. Primary figures 1 to 7 cover schematics, waveforms, and layout. Figures 8 onwards provide supplementary validation: ring architecture (Figure 8), paper sizing waveforms (Figure 9), design B waveforms at 6 GHz (Figure 10), phase steps (Figure 11), tail node voltages (Figures 12, 13, and 14), supply current (Figure 15), frequency sweeps (Figure 16), PVT distributions (Figures 17, 18, and 19), testbench schematic (Figure 20), session log (Figure 21), DLATCH layout (Figure 22), DRC summary (Figure 23), and transient waveform plot for design D (Figure 24).

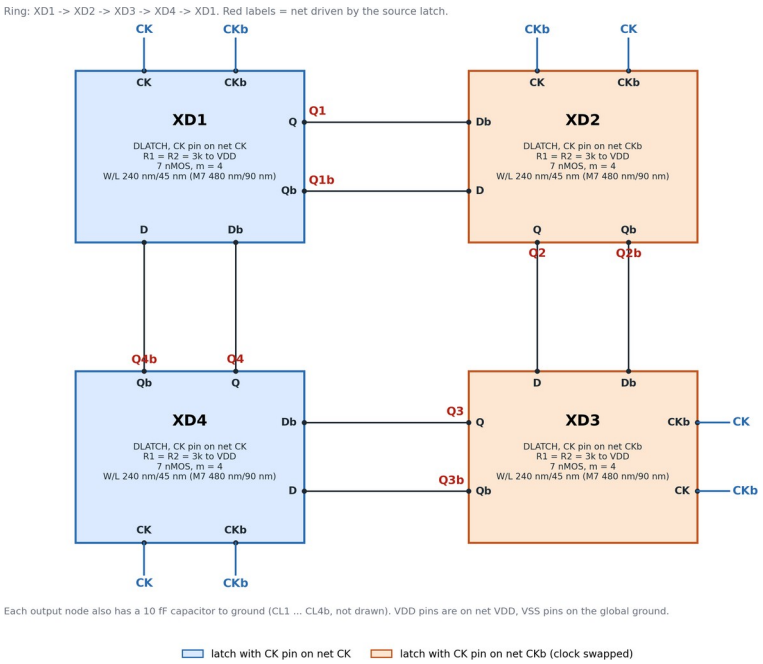


Figure 8: Ring interconnection of four DLATCH stages (XD1 to XD4) with alternating cross-connections, and inverted clocks on XD2 and XD3 (design D: m = 4, R = 3 kOhm).

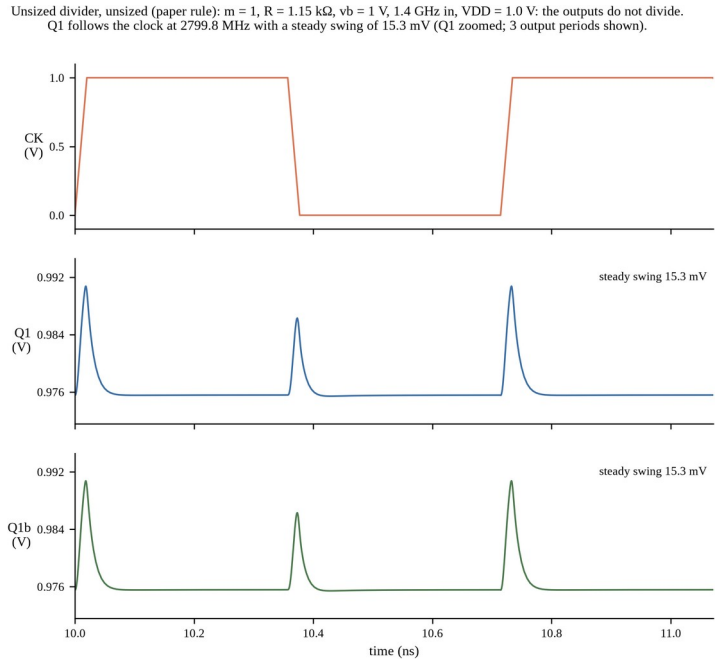


Figure 9: Transient simulation with paper sizing (m = 1, R1 = R2 = 1.15 kOhm = Ron/2) at 1.4 GHz. Division fails: Q1 oscillates at 2799.8 MHz with only 15.3 mV swing (0.975 V to 0.991 V).

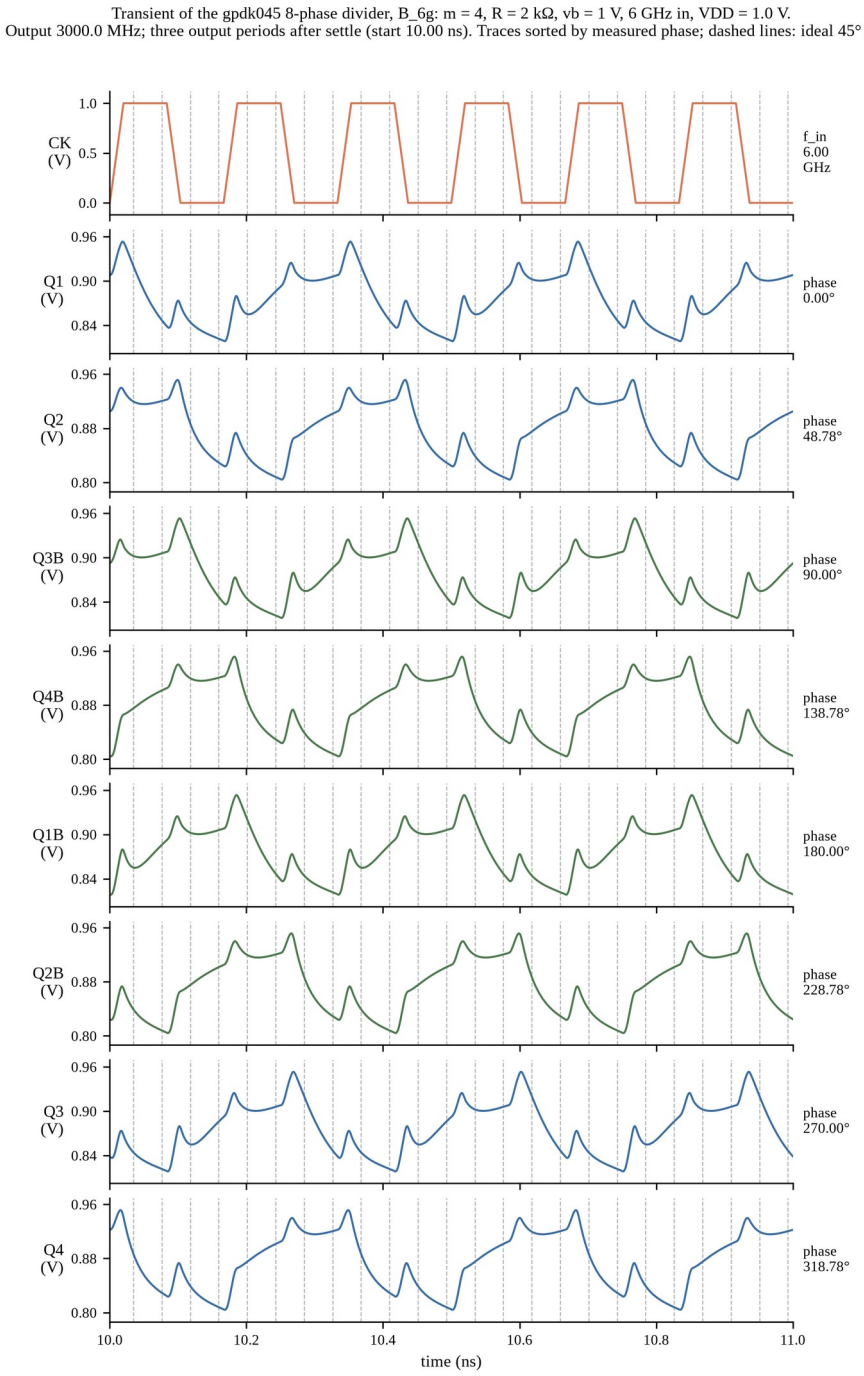


Figure 10: Transient waveforms for design B at 6 GHz (m = 4, R1 = R2 = 2 kOhm), yielding divided outputs at 3000.0 MHz.

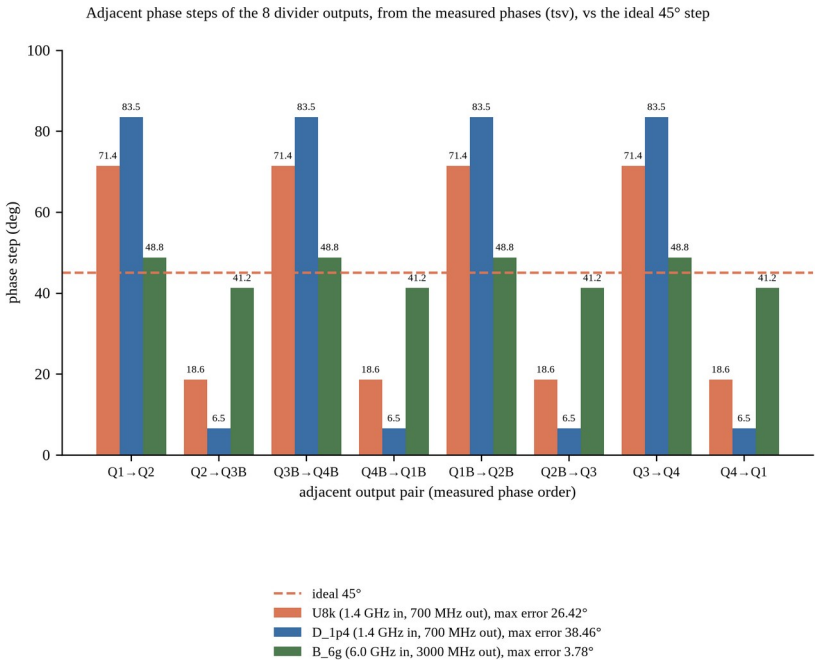


Figure 11: Phase steps between the eight outputs for baseline U8k (1.4 GHz), design D (1.4 GHz), and design B (6 GHz).

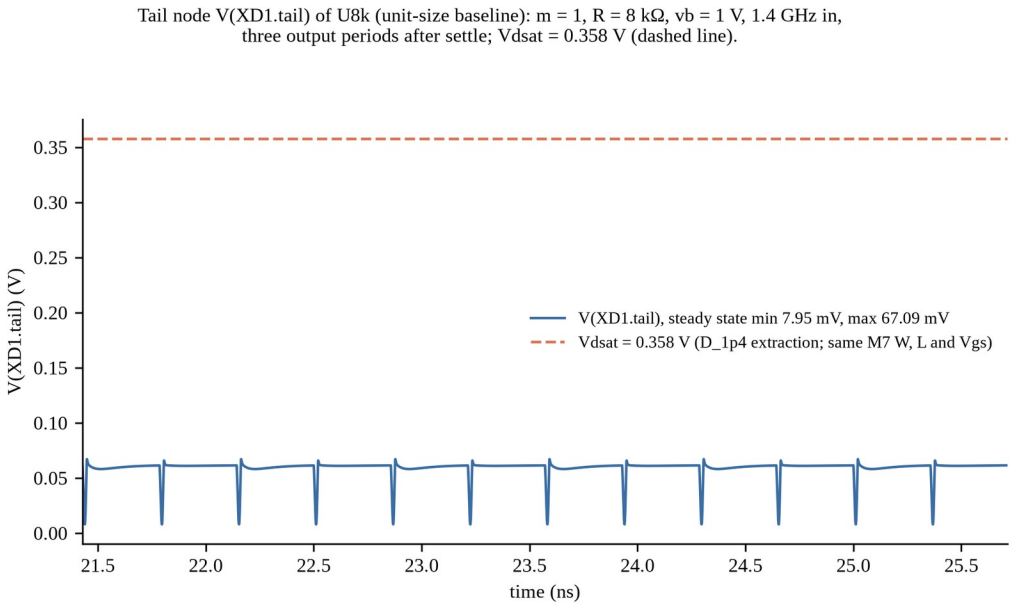


Figure 12: Tail transistor (M7) drain voltage for baseline U8k compared with saturation voltage Vdsat = 0.358 V.

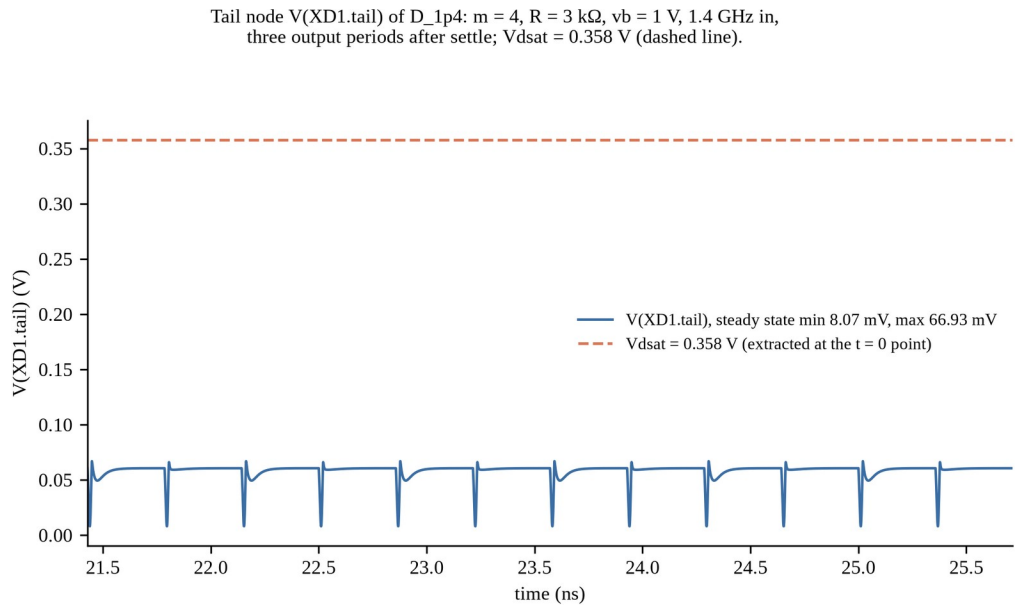


Figure 13: Tail transistor (M7) drain voltage for design D at 1.4 GHz compared with saturation voltage Vdsat = 0.358 V.

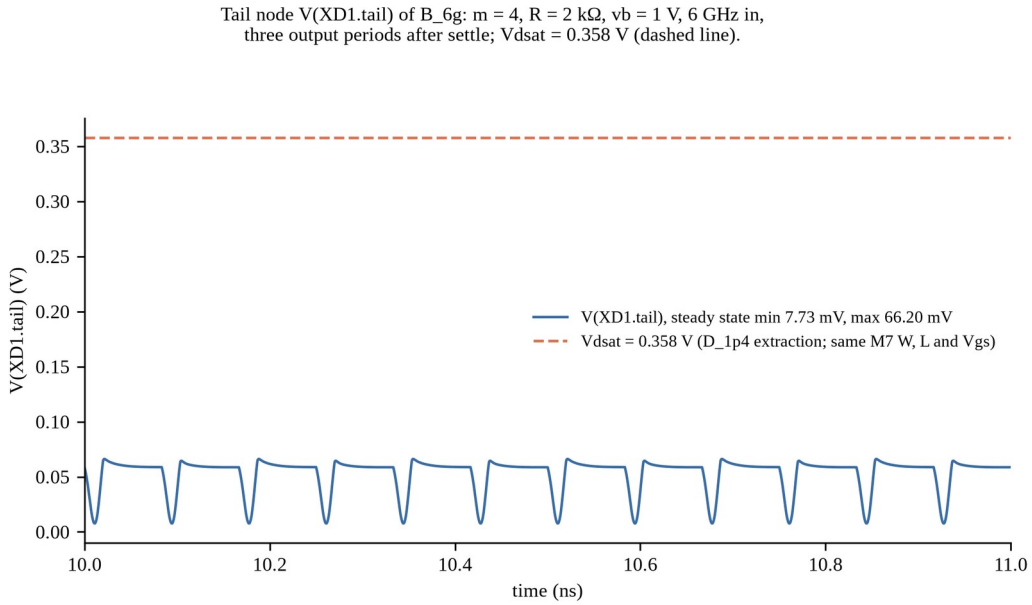


Figure 14: Tail transistor (M7) drain voltage for design B at 6 GHz compared with saturation voltage Vdsat = 0.358 V.

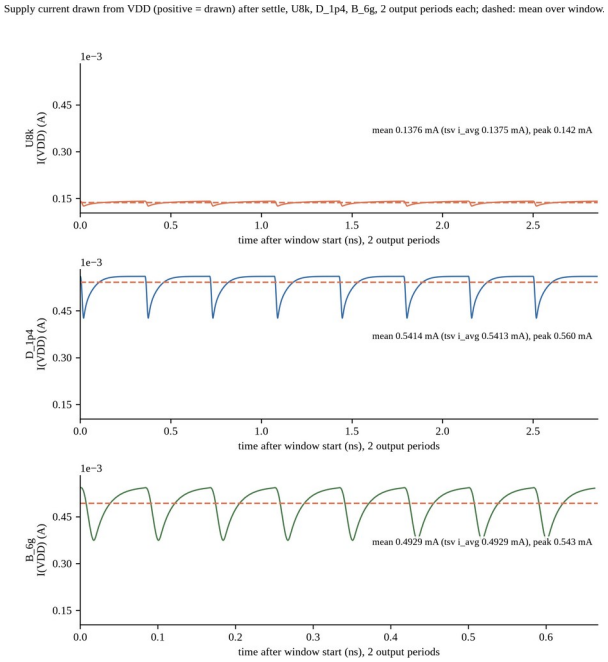


Figure 15: Supply current drawn from VDD for baseline U8k, design D, and design B.

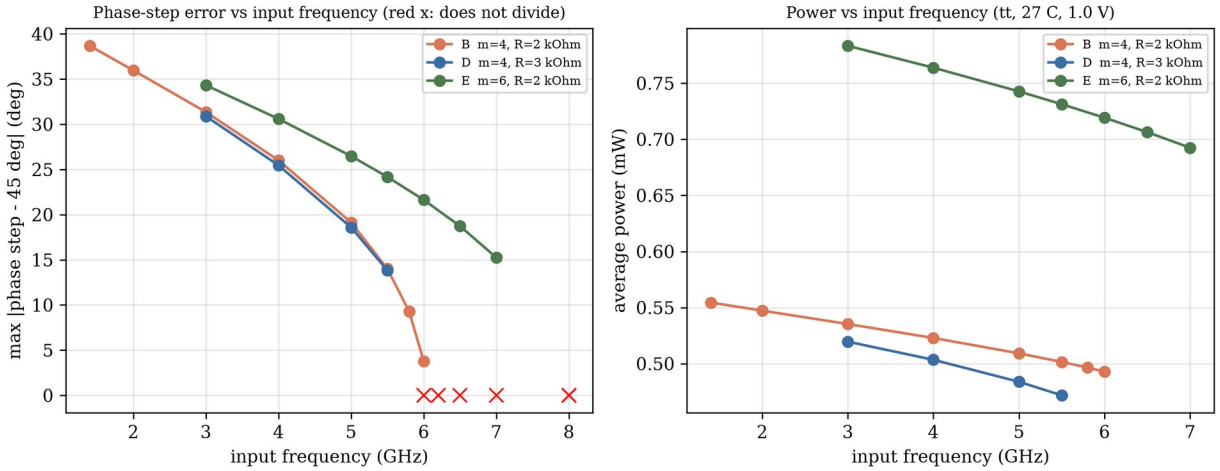


Figure 16: Frequency sweeps for designs B, D, and E showing division range and maximum phase-step error versus input frequency.

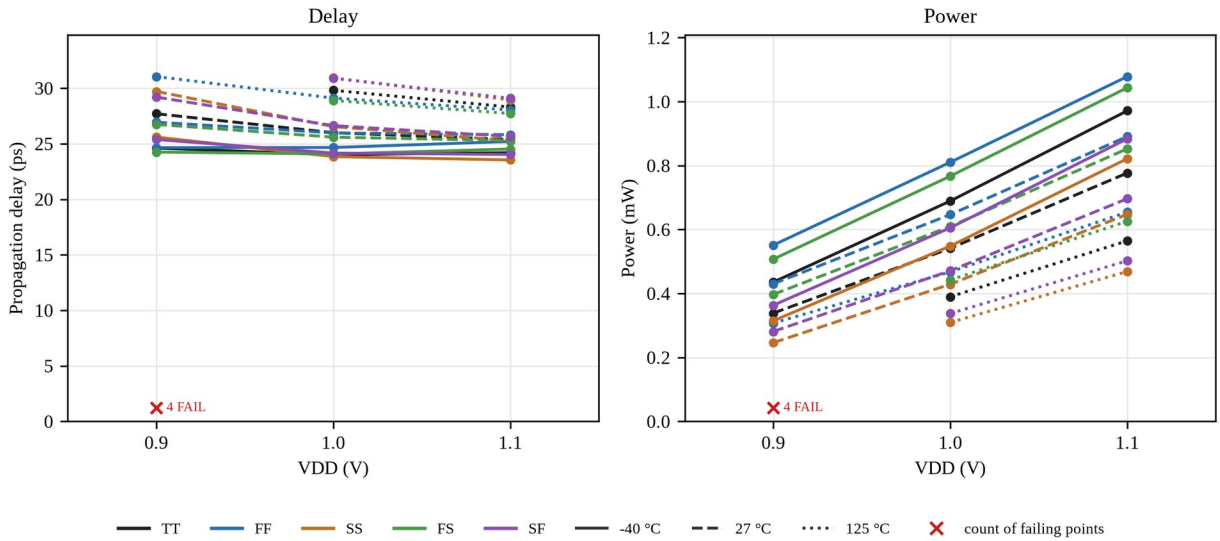


Figure 17: Delay and power across the PVT matrix for design D at 1.4 GHz (crosses mark failing corners).

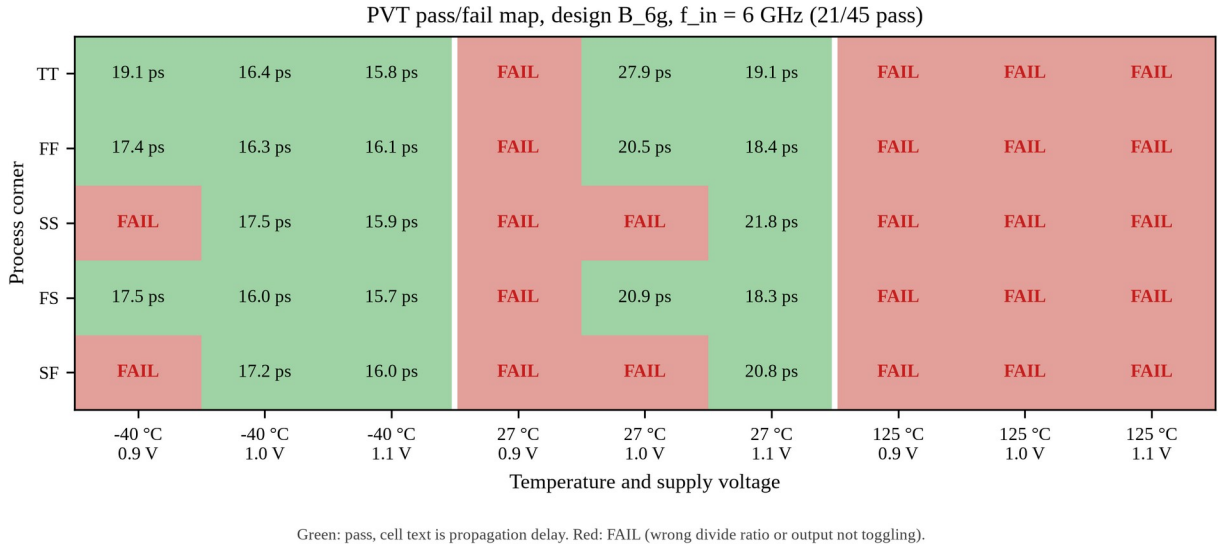


Figure 18: PVT pass/fail map for design B at 6 GHz (21 of 45 points pass).

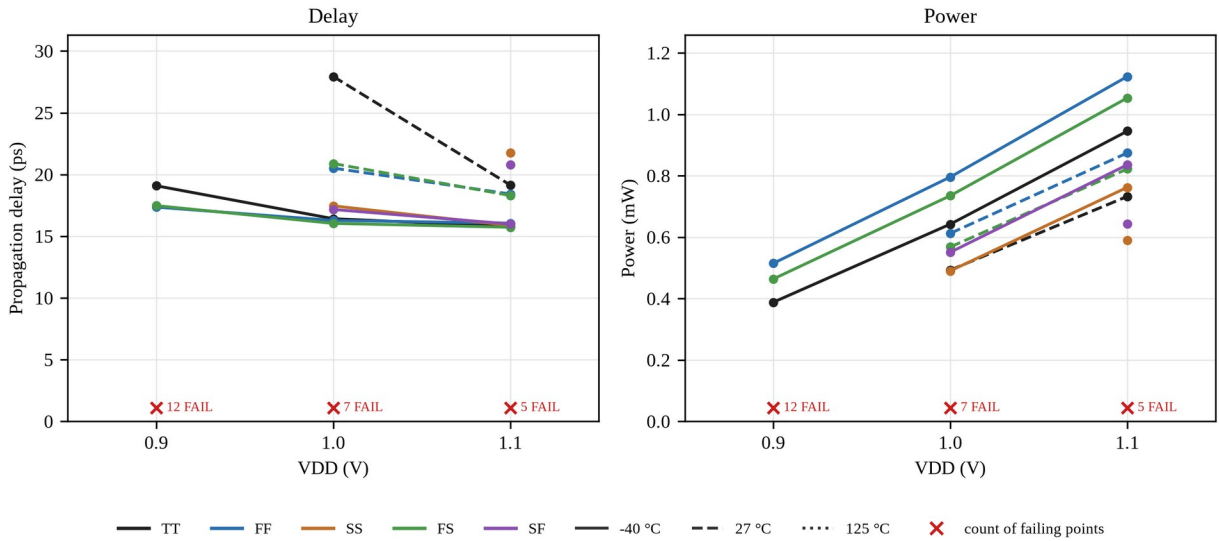


Figure 19: Delay and power across the PVT matrix for design B at 6 GHz (crosses mark failing corners).

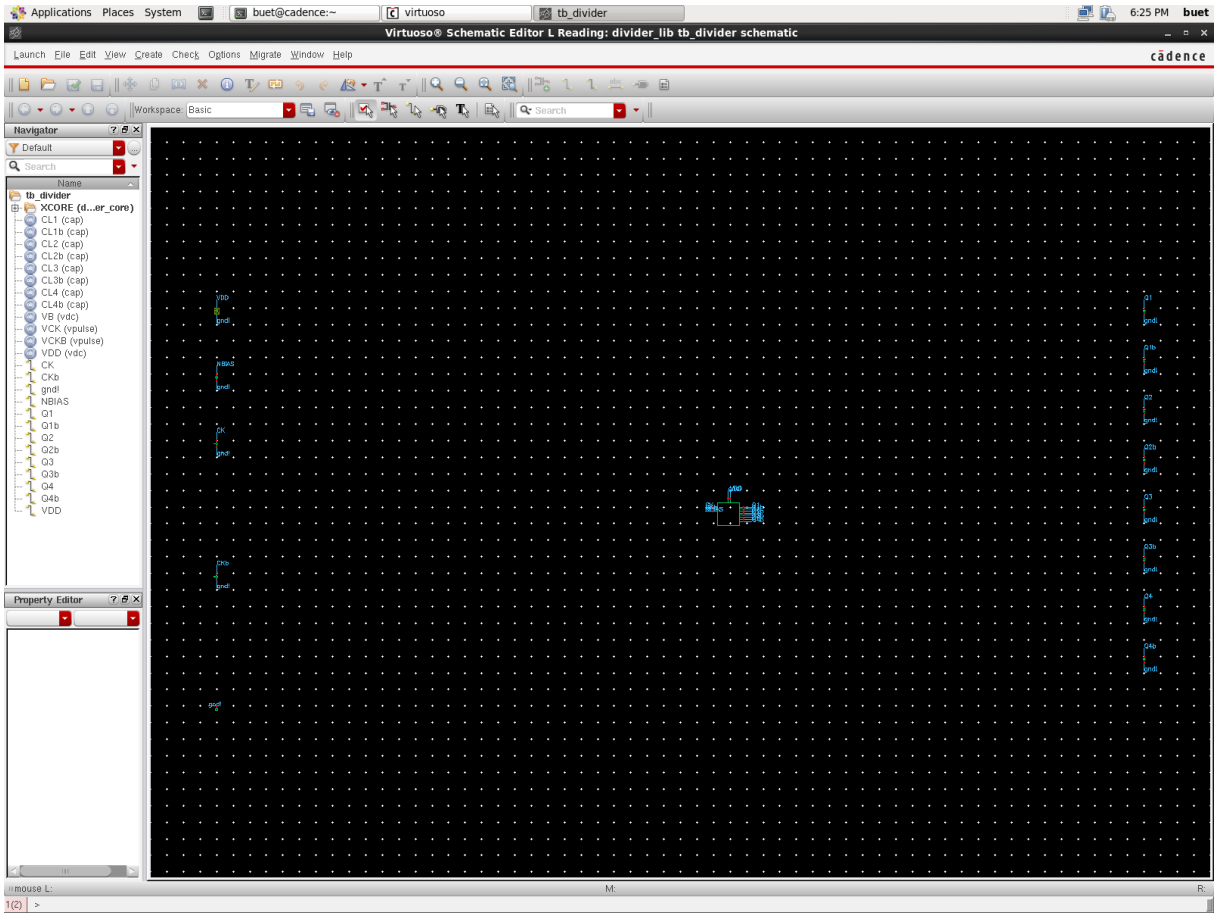


Figure 20: Virtuoso schematic editor window showing testbench tb_divider. Schematic check: DLATCH 0/0, divider_core 0/0, tb_divider 0/1 errors/warnings.

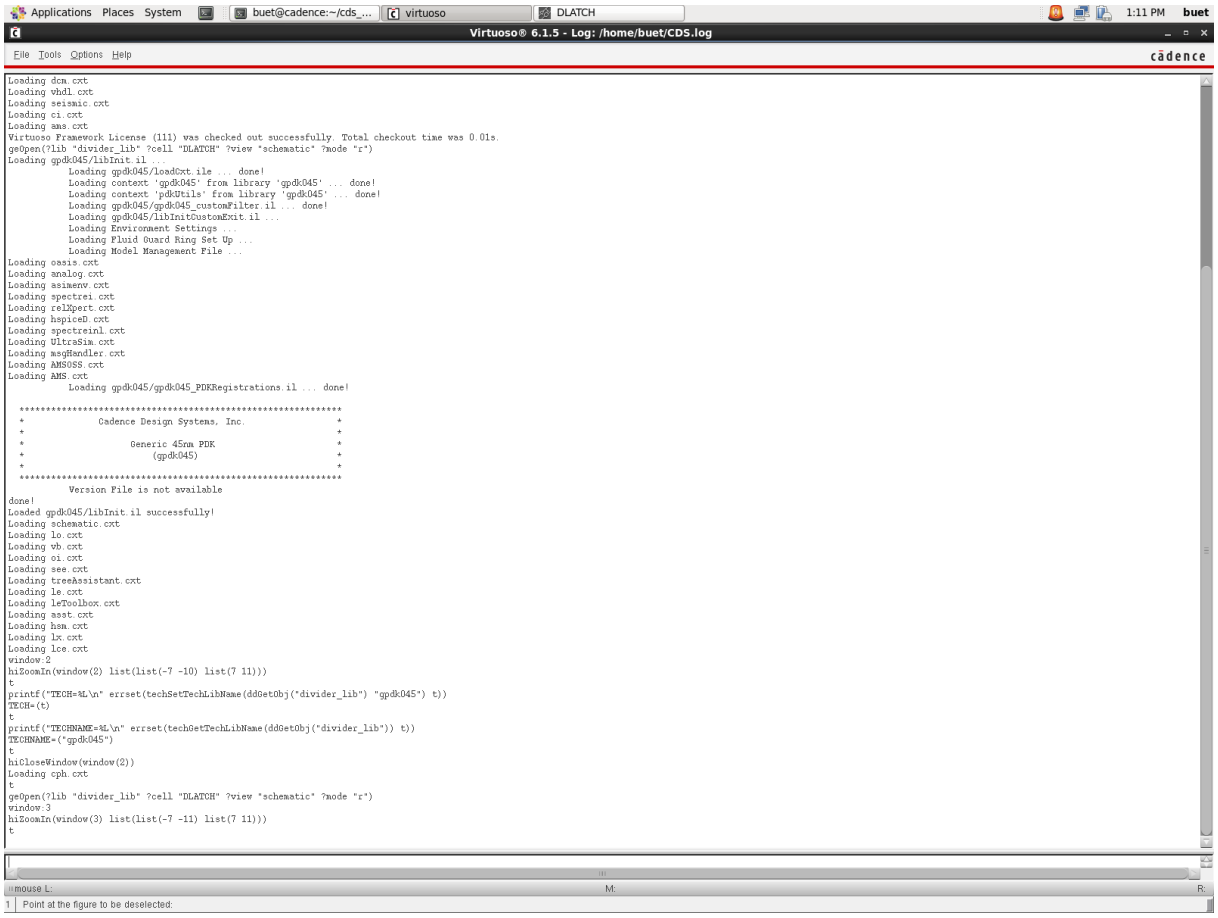


Figure 21: Virtuoso Command Interpreter Window (CIW) log confirming gpdk045 library initialization.

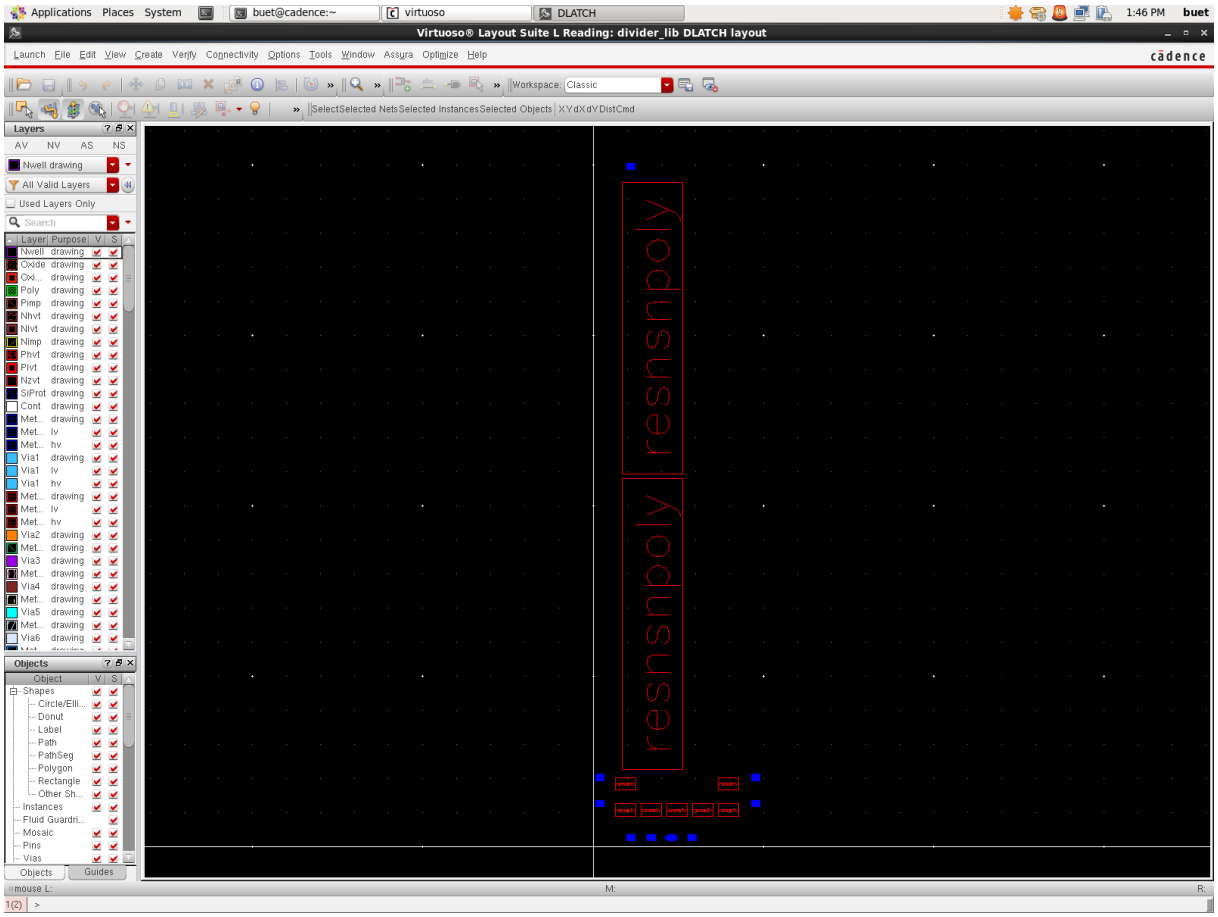


Figure 22: Virtuoso layout editor showing the DLATCH cell: 7 unit nMOS devices (m = 1) and 2 poly resistors (placement only, unrouted).

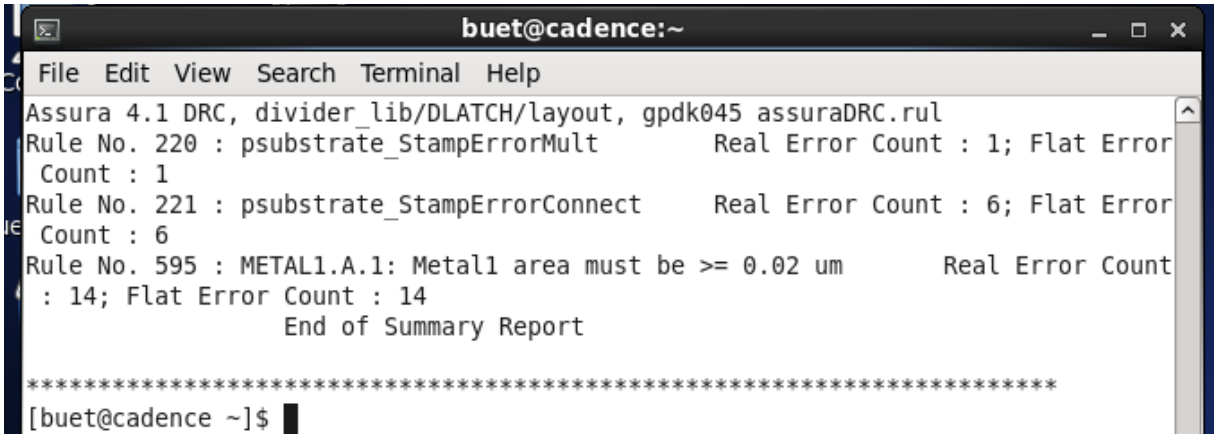


Figure 23: Assura DRC summary for the DLATCH layout showing 21 errors across 3 rules.

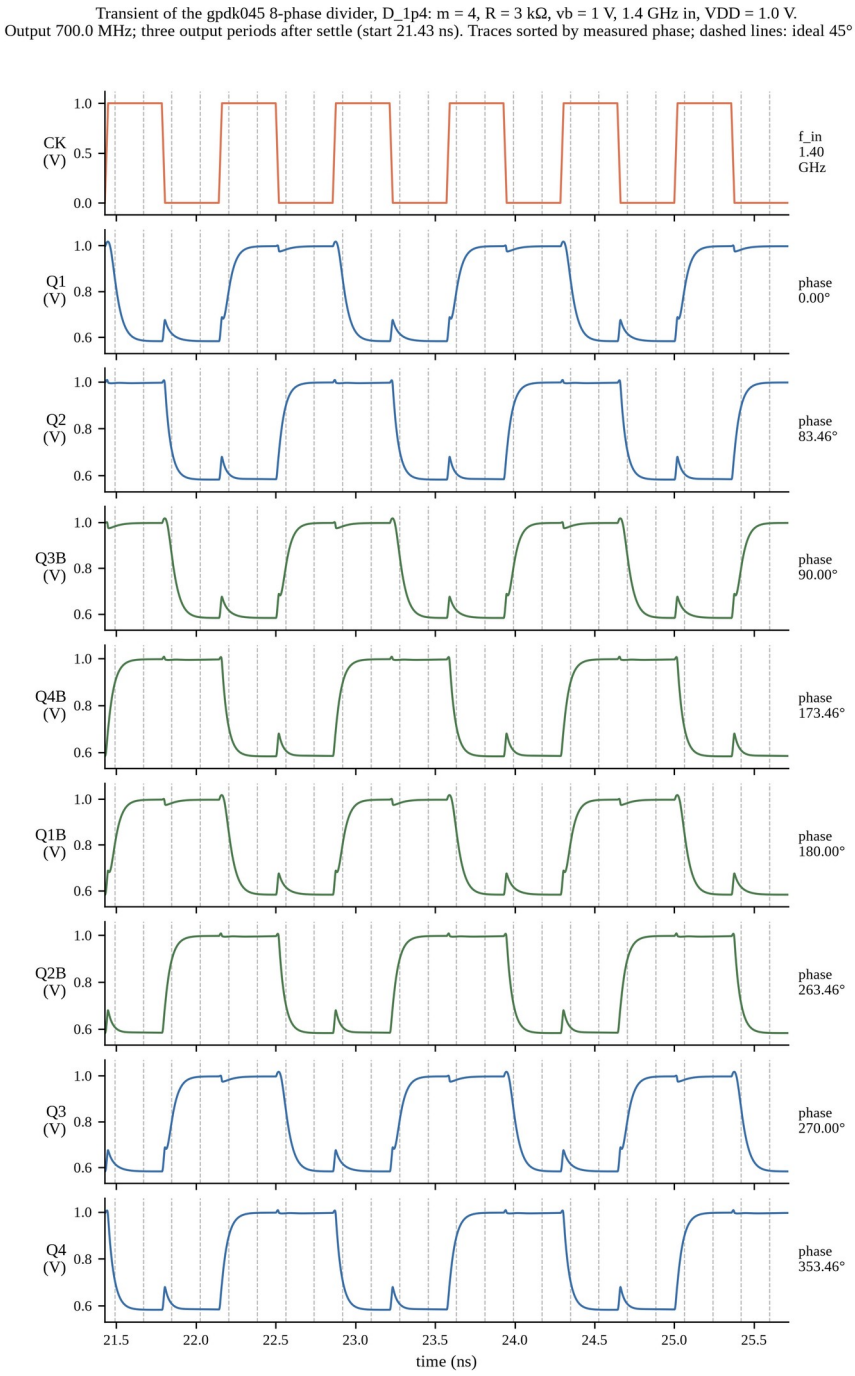


Figure 24: Transient waveforms for design D (m = 4, R1 = R2 = 3 kOhm) at 1.4 GHz. Output frequency is 700.0 MHz with Q1 swinging from 0.583 V to 1.017 V. Tail voltage V(XD1.tail) remains between 8.1 mV and 66.9 mV (XD2 reaches 68.2 mV).

CONCLUSION:

Multiphase CMOS division in 45 nm was demonstrated at two operating points using the stack resistance rule. Design D at 1.4 GHz (m = 4, R = 3 kOhm) achieved 0.5413 mW, 25.99 ps delay, 14.067 fJ PDP, and 38.46 deg phase error (41 of 45 PVT corners passing). Design B at 6 GHz (m

= 4, R = 2 kOhm) achieved 0.4929 mW, 27.94 ps delay, 13.768 fJ PDP, and 3.78 deg phase error (21 of 45 PVT corners passing). Design B exhibited reduced swing (0.135 V, 0.818 V to 0.953 V against the 0.1 V limit), failed at tt, 27 C, 0.9 V, and failed division at 6.2 GHz.

Baseline comparison: sizing cuts delay from 62.21 ps (U8k) to 25.99 ps (D) (2.39x) and increases swing from 0.2289 V to 0.4344 V. Power rises from 0.1375 mW to 0.5413 mW (3.94x), PDP from 8.557 fJ to 14.067 fJ (1.64x), and phase-step error from 26.42 deg to 38.46 deg at 1.4 GHz.

The paper sizing rule (m = 1, R1 = Ron/2 = 1.15 kOhm) fails to divide at 1.4 GHz because load resistance is below the regeneration threshold (7.59 kOhm). Empirical division thresholds span 7 to 8 kOhm at vb = 0.8 V, and 6 to 7 kOhm at vb = 1 V.

Operating limitations: transistor M7 operates in triode across all designs. Phase-step error at 1.4 GHz (38.46 deg) exceeds the literature value (3.75 deg); 3.78 deg was reached only at 6 GHz. All 4 PVT failures for design D occur at 125 C and 0.9 V. Design B operates near the regeneration boundary, passing 21 of 45 PVT points at 6 GHz and 27 of 45 at 1.4 GHz.

Layout implementation: placement was performed for unit-size transistors (m = 1) and placeholder resistors (2000 ohm); routing, LVS, extraction, and post-layout simulation were not performed (design D with m = 4 and R = 3 kOhm was not laid out). DLATCH DRC yields 21 errors across 3 rules (7 substrate connectivity, 14 Metal1 minimum area).

PROBLEMS FACED DURING THE IMPLEMENTATION:

Compilation failure in Spectre AHDL/CMI module compilation for the gpdk045 BSIM4 model.

32-bit simulation environment compatibility with Cadence tool dependencies and compiler toolchain.

Storage space limitations during multi-corner parameter sweeps.

Failure of paper sizing (R1 = Ron/2) to divide at 1.0, 1.4, or 2.0 GHz, requiring an empirical search for load resistance.

False division detection caused by low-amplitude harmonics at 2 f_in (2.8 GHz at 1.4 GHz input) with only 13 to 20 mV swing.

Transient raw data parsing constraints during multi-signal simulation extraction.

High-temperature PVT failure at 125 C for marginal sizing designs.

Data transfer constraints within the isolated simulation environment.

Large phase-step error at 1.4 GHz (38.46 deg) compared with the reference literature value of 3.75 deg.

Layout incomplete: placement only (m = 1) without routing, yielding 21 DRC errors (7 substrate stamp, 14 Metal1 minimum area); LVS and post-layout simulation were not run.

PROBLEMS RESOLVED USING SOLUTIONS:

Configured host C compiler (gcc 4.4.7) and runtime libraries to support Spectre AHDL model compilation.

Extracted and validated simulation parameters systematically across nominal, corner, and sweep conditions.

Identified dividing unit-size load resistance across 12 simulations and applied the stack resistance rule to establish design D (41 of 45 PVT points passing).

Enforced strict division verification: all eight phases must operate at f_in/2 within 0.1% with peak-to-peak swing >= 0.1 V.

Verified consistency across transient output metrics and automated simulation summaries.

Characterized divider performance at two operating points: 1.4 GHz (reference comparison) and 6 GHz (minimized phase error).

Executed batch Assura physical verification on the DLATCH cell to quantify placement design rule violations.

ANY OTHER COMMENT:

Waveform and PVT figures present transient simulation data extracted from Spectre.

On-resistance Ron and division boundaries reflect measured device operating points.

Circuit diagrams reflect netlist connectivity and device parameters for design D.

Reference specifications reflect published literature data (Kuo et al.).

REFERENCES:

Yue-Fang Kuo, Shih-Hung Lin, Bo-Yi Chen and Liang-Chen Liu, "A Compact Multiphase Divide-by-Two CMOS Frequency Divider," IEEE Microw. Wireless Technol. Lett., vol. 35, no. 5, pp. 617-620, May 2025, doi: 10.1109/LMWT.2025.3528863.

Cadence Design Systems, Cadence GPDK045 v6.0 documentation (45 nm generic process design kit).

Cadence Design Systems, Spectre Circuit Simulator Reference, MMSIM 12.1.

Evidence file: results/gpdk045/ocean_schematic_run_1p4GHz_D.txt

Evidence file: cadence/ocean/run_schematic_tb_param.ocn

Evidence file: results/gpdk045/final/D_1p4.tsv

APPENDIX: COMPLETE PVT DATA

Complete 45-corner simulation results for design D at 1.4 GHz across process corners (tt, ff, ss, fs, sf), temperatures (-40, 27, 125 °C), and supply voltages (0.9, 1.0, 1.1 V). Total 41 corners pass division to 700.0 MHz; 4 corners fail (all at 125 °C, 0.9 V for tt, ss, fs, sf).

Table 6: Complete PVT analysis results for design D at 1.4 GHz (45 corners)

Corner	Temp (°C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
tt	-40	0.9	PASS	700.0	24.61	0.4356	10.72	
tt	-40	1.0	PASS	700.0	24.01	0.6892	16.54	
tt	-40	1.1	PASS	700.0	24.19	0.9714	23.50	
tt	27	0.9	PASS	700.0	27.70	0.3387	9.38	
tt	27	1.0	PASS	700.0	25.99	0.5413	14.07	
tt	27	1.1	PASS	700.0	25.38	0.7769	19.72	
tt	125	0.9	**FAIL**	2799.7	n/a	n/a	n/a	outputs at 2 f_in, swing 0.034 V < 0.1 V
tt	125	1.0	PASS	700.0	29.81	0.3898	11.62	
tt	125	1.1	PASS	700.0	28.30	0.5648	15.98	
ff	-40	0.9	PASS	700.0	24.65	0.5512	13.59	
ff	-40	1.0	PASS	700.0	24.66	0.8104	19.99	
ff	-40	1.1	PASS	700.0	25.20	1.0784	27.17	
ff	27	0.9	PASS	700.0	26.94	0.4303	11.59	
ff	27	1.0	PASS	700.0	25.97	0.6467	16.80	
ff	27	1.1	PASS	700.0	25.81	0.8910	23.00	
ff	125	0.9	PASS	700.0	31.05	0.3074	9.55	
ff	125	1.0	PASS	700.0	29.11	0.4677	13.61	
ff	125	1.1	PASS	700.0	28.04	0.6547	18.36	
ss	-40	0.9	PASS	700.0	25.60	0.3147	8.06	
ss	-40	1.0	PASS	700.0	23.84	0.5483	13.07	
ss	-40	1.1	PASS	700.0	23.54	0.8217	19.34	
ss	27	0.9	PASS	700.0	29.71	0.2463	7.32	
ss	27	1.0	PASS	700.0	26.54	0.4288	11.38	

Corner	Temp (°C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
ss	27	1.1	PASS	700.0	25.27	0.6476	16.37	
ss	125	0.9	**FAIL**	2799.7	n/a	n/a	n/a	outputs at 2 f_in, swing 0.029 V < 0.1 V
ss	125	1.0	PASS	700.0	30.93	0.3095	9.57	
ss	125	1.1	PASS	700.0	28.94	0.4688	13.57	
fs	-40	0.9	PASS	700.0	24.25	0.5072	12.30	
fs	-40	1.0	PASS	700.0	24.08	0.7669	18.46	
fs	-40	1.1	PASS	700.0	24.52	1.0436	25.59	
fs	27	0.9	PASS	700.0	26.74	0.3971	10.62	
fs	27	1.0	PASS	700.0	25.59	0.6098	15.61	
fs	27	1.1	PASS	700.0	25.29	0.8525	21.56	
fs	125	0.9	**FAIL**	700.0	n/a	n/a	n/a	divides, swing 0.093 V < 0.1 V
fs	125	1.0	PASS	700.0	28.89	0.4423	12.78	
fs	125	1.1	PASS	700.0	27.72	0.6259	17.35	
sf	-40	0.9	PASS	700.0	25.38	0.3632	9.22	
sf	-40	1.0	PASS	700.0	24.16	0.6055	14.63	
sf	-40	1.1	PASS	700.0	24.03	0.8837	21.23	
sf	27	0.9	PASS	700.0	29.20	0.2811	8.21	
sf	27	1.0	PASS	700.0	26.63	0.4713	12.55	
sf	27	1.1	PASS	700.0	25.65	0.6964	17.86	
sf	125	0.9	**FAIL**	2799.7	n/a	n/a	n/a	outputs at 2 f_in, swing 0.031 V < 0.1 V
sf	125	1.0	PASS	700.0	30.89	0.3375	10.43	
sf	125	1.1	PASS	700.0	29.09	0.5023	14.61	